

ZHUOYUN DU

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🎯 RESEARCH INTERESTS

Focus on **Large Language Models (LLMs)** and **Agentic Systems**. Main research interests include: **Multi-agent collaboration mechanisms**, **Autonomous agent evolution**, and **Agent latent space reasoning and communication**. [🏠 Homepage](#) · [🎓 Google Scholar](#)

🎓 EDUCATION

Zhejiang University, Hangzhou, China Sept 2024 – Jul 2027 (Expected)
M.Eng. in Data Science and Engineering (Recommendation Exemption) Advisor: [Prof. Wei Chen](#)

Jinan University, Guangzhou, China Sept 2020 – Jul 2024
B.Eng. in Computer Science and Technology (Rank: 3 / 99)

👥 PROJECT EXPERIENCE

Multi-Agent Software Development via Cross-Team Collaboration November 2023 – June 2024
Leader

Proposed a novel multi-agent collaborative framework that effectively organizes and schedules multiple customizable generative agent teams. By introducing collaborative interactions between teams at key stages, this approach successfully breaks the closed nature of individual agent teams. Responsible for the entire project lifecycle, including conceptualization, design, and coding. Paper accepted by ACL 2025. [\[Croto\]](#), [\[Project page\]](#).

Agent Coevolution Framework for Standardized Patients August 2024 – December 2024
Leader

Proposed a coevolutionary framework for standardized patient agents. By collecting autonomous dialogue data between doctor and patient agents, it achieves mutual enhancement in interaction quality. This method improves agent evolution efficiency and significantly strengthens the patient agent's adherence to standardized settings, making it effective for training human doctors. Paper accepted by ACL 2025. [\[EvoPatient\]](#), [\[Project page\]](#).

Enabling Agents to Communicate Entirely in Latent Space January 2025 – September 2025
Leader

Innovatively proposed a new paradigm for direct agent communication in latent space. By transmitting hidden states instead of traditional natural language, combined with efficient communication compression methods, it significantly enhances the performance and communication efficiency of multi-agent systems in complex tasks. Paper accepted by ACL 2026. [\[Interlat\]](#), [\[Project page\]](#).

Multi-Agent Collaboration Network November 2023 – June 2024
Member

Deployed LLM agents based on a Directed Acyclic Graph (DAG) topology, transforming complex networks into routing sequences for agent interaction via topological sorting, and driving task-oriented collaboration with natural language interactions. Primarily responsible for developing the agent information integration module, profile generation, and drafting parts of the paper. Paper accepted by ICLR 2025. [\[MACNet\]](#), [\[Project page\]](#).

💼 WORK EXPERIENCE

ByteDance Data Dept Jindouyun Talent Program Intern March 2026 – Present

- **Core Research (LLM Post-training, Preference Alignment & Interactive Evaluation):** Designed and built an automated pipeline based on high-fidelity human-like agents. Simulated diverse user profiles and behavioral patterns to generate high-quality data supporting SFT and RLHF/DPO processes, and constructed dynamic interactive evaluation mechanisms to assess model multi-turn context processing and preference alignment capabilities.

Alibaba Future Life Lab Research Intern January 2025 – March 2026

- **Core Research:** Focused on multi-agent reasoning and communication in latent space.
- **Project Leadership:** Led a 9-month frontier research project "Agent Communication Entirely in Latent Space", replacing traditional natural language with hidden states and leveraging efficient communication

compression to significantly boost multi-agent performance and efficiency in complex tasks.

- **Engineering Practice:** Gained extensive experience in LLM fine-tuning and large-scale distributed multi-GPU (128+) training; authored two first-author papers during the internship ([Interlat](#) & [Online-PVLM](#)).

Tsinghua University THUNLP Research Assistant December 2023 – August 2024

- **Swarm Intelligence Research:** Participated in multi-agent swarm intelligence research based on [ChatDev](#) in Prof. [Zhiyuan Liu](#)'s group, exploring the enhancement of emergent collective capabilities through cooperation and competition mechanisms among LLMs.
- **Open Source Contribution:** Core contributor/leader for key branches ([MACNet/Croto](#)), with related open-source projects accumulating over 32k stars on GitHub.

🔧 RELATED SKILLS

- **LLM & Deep Learning:** Multi-Agent System architecture design, cluster-level distributed multi-GPU training experience (128+ GPUs); proficient in SFT, LoRA and other fine-tuning paradigms; familiar with training acceleration technologies like DeepSpeed and Flash Attention.
- **Programming Languages & Frameworks:** Proficient in Python, C/C++, Bash, Git, $\text{\LaTeX}$; heavily rely on PyTorch and Hugging Face ecosystem (Transformers, Datasets) for algorithm development.
- **Languages:** English (Fluent, IELTS: 7.5, CET-6: 598); Mandarin (Native).

🏆 AWARDS AND HONORS

- 2025 **National Scholarship**, Outstanding Graduate Student Scholarship, **3 Undergraduate Scholarships**, **First-Class Scholarship for Outstanding Graduates**.

🔥 INTERESTS AND SPECIALTIES

Broad academic vision and strong intellectual curiosity, with excellent collaboration and leadership skills in complex cross-team cooperation. Diverse hobbies, former **National Champion in Amateur Épée Fencing**, hold a Grade 7 Piano Certificate, and passionate about snowboarding, photography, and various arts and cultural activities.

📖 PUBLICATIONS

1. **Zhuoyun Du**[†], Chen Qian[†], Wei Liu, Zihao Xie, Yifei Wang, Yufan Dang, Weize Chen, and Cheng Yang. Multi-agent software development through cross-team collaboration. *Findings of ACL 2025*.
2. **Zhuoyun Du**[†], Lujie Zheng[†], Renjun Hu, Yuyang Xu, Xiawei Li, Ying Sun, Wei Chen, Jian Wu, Hao lei Cai, and Hao hao Ying. LLMs can simulate standardized patients via agent coevolution. *ACL 2025*.
3. **Zhuoyun Du**[†], Runze Wang[†], Huiyu Bai, Zouying Cao, Xiaoyong Zhu, Bo Zheng, Wei Chen and Haochao Ying. Enabling Agents to Communicate Entirely in Latent Space. *ACL 2026*.
4. Huiyu Bai[†], Runze Wang[†], **Zhuoyun Du**[†], Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng and Xuejiao Zhao. Online-PVLM: Advancing Personalized VLMs with Online Concept Learning. *arXiv preprint arXiv:2511.20056*, 2025.
5. Shan He, Runze Wang, **Zhuoyun Du**, Huiyu BAI, zouying cao, YuCheng, Bo Zheng. Learning to Evolve: A Self-Improving Framework for Multi-Agent Systems via Textual Parameter Graph Optimization. *Findings of ACL 2026*.
6. Chen Qian[†], Zihao Xie[†], Yifei Wang[†], Wei Liu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, Zhiyuan Liu, and Maosong Sun. Scaling large-language-model-based multi-agent collaboration. *ICLR 2025*.
7. Wei Liu[†], Chenxi Wang[†], Yifei Wang, Zihao Xie, Rennai Qiu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, and Chen Qian. Autonomous agents for collaborative task under information asymmetry. *NeurIPS 2024*.
8. Yuyang Xu, Yi Cheng, Haochao Ying, **Zhuoyun Du**, Renjun Hu, Xing Shi, Wei Lin and Jian Wu. SSPO: Self-traced Step-wise Preference Optimization for Process Supervision and Reasoning Compression. *arXiv preprint arXiv:2508.12604*, 2025.

[†] equal contribution.